GreenProof

A greenwashing screener for corporate sustainability reports – project overview for our group

In one sentence: We build a tool that reads sustainability reports, finds every environmental claim and checks whether it would hold up under the new EU green-claims rules – and whether the company backs it up with real climate data. So we can compare how honest companies really are.

1. What is greenwashing?

Greenwashing means giving a misleading impression of how environmentally friendly a company, product or service is. It is rarely an outright lie. Usually it is a claim that sounds good but cannot be measured, leaves out the important part, or relies on someone else's effort. Typical patterns:

- **Vague words** – "eco-friendly", "green", "sustainable" without any number or proof.
- **Offsetting** – calling a product "carbon neutral" because the company bought carbon credits, not because it cut its own emissions.
- **Overstating** – "zero impact" or "100% sustainable" when only one part of the product improved.
- **Empty promises** – "net zero by 2040" without a baseline, interim targets or a plan.

Real cases

• **KLM (2024):** The District Court of Amsterdam ruled on 20 March 2024 that 15 of 19 challenged statements in KLM's "Fly Responsibly" campaign were misleading – e.g. vague sustainability claims and an overly rosy picture of offsetting and sustainable aviation fuel.

• **Apple (2025):** The Frankfurt Regional Court stopped Apple from advertising three Apple Watch models as "CO_2-neutral". The neutrality relied on offset projects; one plantation in Paraguay was only leased until 2029. (Ruling not yet final when reported in August 2025.)

2. Why this project – and why now?

New EU rules since 27 September 2026. The Directive "Empowering Consumers for the Green Transition" (EU 2024/825) now applies in all EU member states. Among other things it:

- bans claims that a product has a neutral, reduced or positive climate impact **based on offsetting**;
- bans **generic environmental claims** that cannot be backed by recognised excellent environmental performance;
- bans claims that present a benefit for the **whole product or company** when it applies only to a part;
- requires **future performance claims** (e.g. "net zero by 2040") to rest on a detailed, realistic implementation plan that is verified by an independent expert.

Nobody checks systematically. A second law that would have required companies to verify green claims before publishing them (the Green Claims Directive) stalled in June 2025. Research also shows the problem is real: Bingler et al. (2022) found that much corporate climate disclosure is "cheap talk" – many words, little measurable substance.

Our contribution: a transparent, repeatable screening method that turns hundreds of report sentences into a comparable score – something investors, auditors and consumers cannot easily do by hand.

3. How does this relate to ESG?

ESG pillar	Link to our project
E – Environment	We assess the quality of environmental disclosure: are claims about emissions, energy, materials and packaging specific and verifiable?
G – Governance	Honest reporting is a governance issue: misleading claims create legal and reputational risk and show weak internal control over what a company publishes.
Investors & ESG data	ESG ratings and sustainable funds rely heavily on what companies report about themselves. If claims are inflated, capital flows to the wrong companies.
Data analytics	The course focus: we turn unstructured text (PDF reports) into structured, comparable data with rules and AI – and measure how reliable that is.

4. What we build: the GreenProof screener

1. Upload Sustainability reports (PDF) → **2. Extract** Sentences with page numbers → **3. Detect** Which sentences are environmental claims? → **4. Rate** One of 9 categories per claim → **5. Compare** Score 0–100 and ranking

Each claim gets one category (first match wins). Green = 100 points, yellow = 50, red = 0; the **Claim Quality Score** is the average per company. The tool always shows *why* a sentence was flagged (the decisive words), so every rating can be checked. This is the **Claim Quality Score** (talk).

	Category	Example
Red	Offset-based neutrality	"This watch is carbon neutral thanks to carbon credits."
Red	Generic claim without proof	"Our collection is eco-friendly and kind to the planet."
Yellow	Overstated / absolute claim	"All our products have zero environmental impact."
Yellow	Neutrality claim, basis unclear	"Our offices are carbon neutral."
Yellow	Future target without plan	"We aim to become net zero by 2040."
Yellow	Vague commitment	"Sustainability is at the heart of everything we do."
Green	Future target with plan	"-50% Scope 1+2 by 2030 vs. 2019, validated by the SBTi."
Green	Specific, measurable claim	"78% of our factory electricity was renewable in 2025."
Grey	Not rated (context)	"Climate change may increase extreme weather events."

Second layer – Evidence Score (0–100). Wording alone is not enough, so the tool also checks whether the report discloses the climate data that would back the claims up. Ten items, 10 points each, each with the page where it was found:

Evidence item	Reference	Evidence item	Reference
Scope 1 emissions	GRI 305-1	Science-based / validated target	GRI 102 · SBTi
Scope 2 emissions	GRI 305-2	Climate transition plan	GRI 102 · EU 2024/825
Scope 3 emissions	GRI 305-3	Carbon credits shown separately	GRI 102
Emission intensity	GRI 305-4	External assurance of GHG data	EU 2024/825
Base year for reductions	GRI 305-5	GRI content index	GRI 1

Talk vs. evidence. Both scores place every company in a simple matrix – the core visual of our results:

	Evidence < 50	**Evidence ≥ 50**
Claim quality ≥ 50	Careful claims, thin data	**Credible**
Claim quality < 50	**Greenwashing risk**	Good data, weak claims

Three modes: (1) transparent keyword rules – free and offline; (2) detection with the published research model ClimateBERT (Stammbach et al., 2023); (3) detection and rating by an AI model (Claude API). Comparing the modes is itself a result for our report. A working version (v0.2) already exists: Python + Streamlit dashboard with upload, ranking, talk-vs-evidence matrix, company detail, built-in hand validation and CSV export.

5. Research question and method

Research question: How do large companies differ in the quality of their environmental claims – and how many of these claims would be problematic under the EU rules that apply from September 2026?

- **Sample:** 5–10 companies from one industry (e.g. sportswear or consumer electronics), latest sustainability report each – so the comparison is fair.
- **Screening:** run all reports through GreenProof; compare Claim Quality, Evidence and red-flag shares.
- **Validation:** each of us labels the same 50–100 claims by hand (blind, directly in the app); we report accuracy and Cohen's kappa, interpreted with Landis & Koch (1977).
- **Deep dive:** for 2–3 companies we check selected claims against reported numbers (e.g. emissions trend vs. target).

Limits we state openly: the EU rules target consumer marketing, so we use them as a benchmark for claim quality, not as a legal check. The tool judges *how* claims are worded, not whether the numbers are true; text in charts and images is not read.

6. How it fits our course deliverables

Course deliverable	What we show
Case study (25 + 10 min)	One company from our sample in depth: background, business model canvas, its green claims – e.g. Apple and the Apple Watch case.
Midterm project (10 min)	Introduction: background, motivation, research question, expected contributions; first screening results.
Final project (20 min)	Full study: literature, method, results, conclusions – with product name, system architecture and a **live demo** of GreenProof.

7. Next steps

- Agree on the industry and the 5–10 companies; collect their latest sustainability reports (PDF).
- Run the reports through GreenProof and fine-tune the rules on real documents.
- Split roles: literature & background · data collection · tool & validation · case study & slides.
- Optional: get an Anthropic API key to compare the rule-based mode with the AI mode.

Sources

Directive (EU) 2024/825 of 28 February 2024 (Empowering Consumers for the Green Transition), applies from 27 September 2026. · Climate Focus (2026): Beyond "carbon neutral" – new EU rules for green claims apply from September 2026. · Osborne Clarke (2024): District Court Amsterdam declares green claims of KLM misleading and unlawful. · ZDFheute (26 Aug 2025): Gericht untersagt Apple-Watch-Werbung. · Packaging Europe (Jun 2025): Green Claims Directive grinds to a halt. · Global Reporting Initiative (2016). GRI 305: Emissions; (2025). GRI 102: Climate Change. · Landis, J. R., & Koch, G. G. (1977). The measurement of observer agreement for categorical data. *Biometrics, 33*(1), 159–174. · Bingler, J. A., Kraus, M., Leippold, M., & Webersinke, N. (2022). Cheap talk and cherry-picking: What ClimateBert has to say on corporate climate risk disclosures. *Finance Research Letters, 47*. · Stammbach, D., Webersinke, N., Bingler, J. A., Kraus, M., & Leippold, M. (2023). Environmental claim detection. *Proceedings of ACL 2023*.